

My Kitchen A Multi-Agent Home-Chef Marketplace

A 0 → 1 product built as a set of purpose-built agents —
each scoped to a single job in the order lifecycle, each
designed to fail gracefully.

The Problem

Home-chef marketplaces sit at an unusually high-stakes intersection: food safety, real-time logistics, and deeply variable, unstructured input — a chef describing their menu by voice or text, in their own words, rather than filling out a rigid form. Most platforms solve this with manual moderation and structured onboarding forms, which makes them slow to onboard new chefs and brittle at scale.

That gap shows up as a set of distinct, unmet jobs to be done:

- **Chefs** need a way to list what they're cooking without translating their own natural description into a platform's rigid schema
- **Customers** need certainty that a listing is actually safe for their allergies or dietary restrictions — not just a best-effort filter
- **Both sides** need checkout to reflect real-time kitchen capacity, not a stale availability calendar that leads to overbooked or cancelled orders
- **Operations** needs orders routed and batched sensibly across chefs, rather than assigned in a way that overloads one kitchen and starves another
- **The platform** needs a way to turn raw, unstructured customer feedback into signal that actually improves the next order — not just a star rating that goes nowhere

The originating insight: a pattern noticed in how home chefs actually describe their menus — free-flowing voice notes and casual text mixing ingredients, quantities, and availability in whatever order came to mind, never in the clean structured fields a listing form expects. That mismatch between how chefs naturally communicate and how a database wants data entered is what pointed to needing an agent to do the translation, rather than a better form.

The Users

Three groups sit on either side of the marketplace, each with a distinct job to be done:

The Home Chef	The Safety-Conscious Customer	The Time-Pressed Local Buyer
SIDE Supply	SITUATION Has a real allergy or dietary restriction	SITUATION Wants a reliable order, not a maybe
JOB TO BE DONE List what I'm actually cooking today, in my own words	JOB TO BE DONE Trust a home-kitchen order over a restaurant's	JOB TO BE DONE Get what was promised, when it was promised
NEED The platform does the structuring work, not me	NEED A hard guarantee, not a best-effort filter	NEED A kitchen with real capacity to fulfill the order

The Approach — Prioritizing by Impact

Reasoning through reach and severity across the core problems:

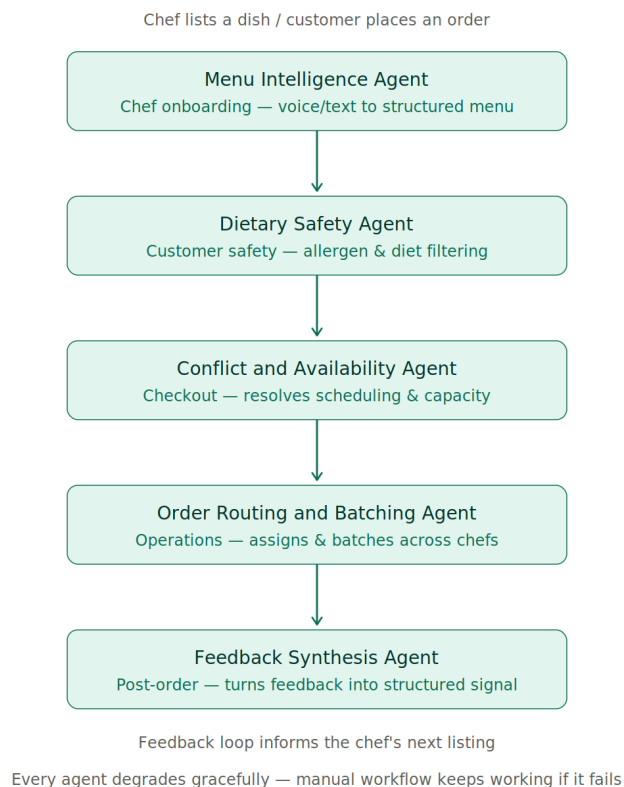
Rank	Problem	Reach	Severity
1	Dietary safety isn't guaranteed	Every safety-conscious customer	High — a single miss is a health incident
2	Onboarding requires structured input	Every new chef joining	High — directly gates supply-side growth
3	Checkout ignores real kitchen capacity	Every order at checkout	High — root cause of overbooking/cancellations
4	Orders aren't routed/batched intelligently	All orders once volume grows	Medium-high — compounding ops cost
5	Feedback doesn't become structured signal	Repeat customers & chefs	Medium — a longer-term quality lever

Strategic note: dietary safety and onboarding friction rank highest because they're trust and supply gates — an unsafe order or a chef who gives up during onboarding ends the relationship before the marketplace ever gets to prove its value.

Solutions for the Top Problems

Rather than one general-purpose assistant, My Kitchen is architected as a set of purpose-built agents, each scoped to a single job in the order lifecycle — from chef onboarding through post-order feedback. Five are built out in depth, mapped directly to the top-ranked problems:

- **Dietary Safety Agent** (customer safety) — filters listings and orders against allergens and dietary restrictions, so safety is enforced structurally rather than left to a chef's best-effort menu description
- **Menu Intelligence Agent** (chef onboarding) — converts voice or free-text descriptions into structured, searchable menu data
- **Conflict and Availability Agent** (checkout) — resolves scheduling and capacity conflicts before an order confirms
- **Order Routing and Batching Agent** (operations) — assigns and batches orders across chefs to avoid overloading one kitchen while another sits idle
- **Feedback Synthesis Agent** (post-order) — turns raw customer feedback into structured signal that can actually improve the next listing



Design principle: graceful degradation. Every agent is designed so that if the AI fails, the underlying manual workflow still works — a core design constraint applied from day one, because a marketplace that breaks when a model has a bad day isn't a marketplace people can trust.

Prioritizing Solutions — Impact vs. Effort

Do First

- Dietary Safety Agent — a rules-plus-model filter on structured allergen data is more tractable to ship correctly than open-ended menu parsing, and directly protects trust

Big Bets

- Menu Intelligence Agent — voice/free-text-to-structured-menu is a harder NLP problem
- Conflict and Availability Agent — real-time capacity resolution across many chefs is a harder systems problem

Fill-Ins

- Order Routing and Batching Agent — meaningfully improves operations cost, but depends on the other agents' data being reliable first

Reconsider

- A fully autonomous Feedback Synthesis loop that auto-edits chef listings without human review — graceful degradation argues for a human-in-the-loop version first, until the agent's reliability is proven

Metrics to Measure Success

Rather than ship agents and hope, each agent is evaluated before expanding scope, with metrics tracked per agent:

Dietary Safety Agent — false-negative rate on flagged allergens (the number that matters most, since a miss here is a safety incident)

Menu Intelligence Agent — % of chef onboarding sessions completed without manual staff intervention

Conflict and Availability Agent — % reduction in post-checkout order cancellations due to capacity conflicts

Order Routing and Batching Agent — average kitchen utilization variance across active chefs

Feedback Synthesis Agent — % of structured feedback signals that lead to a measurable listing change

A real constraint, and what it taught me: early in the design, I explored a WhatsApp group-based architecture for chef-customer coordination — closer to how home-chef networks actually operate informally. That surfaced a hard constraint: WhatsApp's Business API doesn't support the kind of group monitoring this would require without relying on unofficial libraries, which isn't viable for a production system. I dropped the WhatsApp idea and designed a native in-app coordination flow instead — keeping the same informal, low-friction feel chefs were used to, but built on infrastructure the platform actually owns and can monitor for safety and reliability.

Every agent in this system is scoped to one job, evaluated on its own, and built to fail without breaking the marketplace around it.